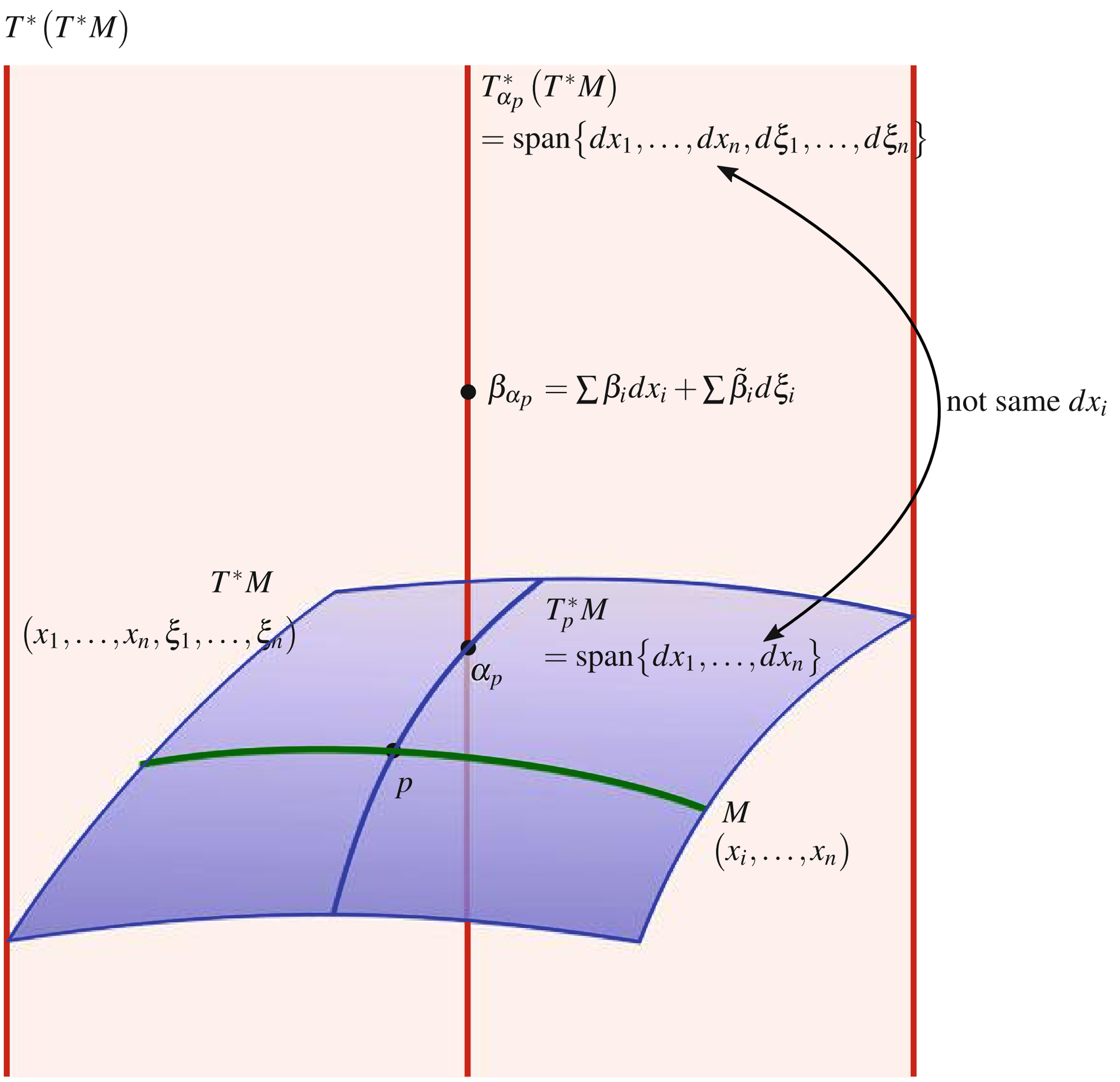

$T^*\left(T^*M\right)$
$T^*_{\alpha_p}\left(T^*M\right)$
$= \mathrm{span}\left\{dx_1, \ldots, dx_n, d\xi_1, \ldots, d\xi_n\right\}$
$\beta_{\alpha_p} = \sum \beta_i dx_i + \sum \tilde{\beta}_i d\xi_i$
not same dx_i
T^*M
$\left(x_1, \ldots, x_n, \xi_1, \ldots, \xi_n\right)$
T^*_pM
$= \mathrm{span}\left\{dx_1, \ldots, dx_n\right\}$
α_p
p
M
$\left(x_i, \ldots, x_n\right)$